

COMP364 Quiz 6

Question 1. Transformer (5 points)

List the major quantities and components of a transformer neural network architecture, in their order of computation.

Solution.

token embedding, position embedding, input representation, a sequence of attention layers, logits via unembedding, token probabilities via softmax, token prediction.

See ?? for some optional additional explanation of this computation.

Question 2. Transformer hyper-parameter (5 points)

Name and describe the most fundamental hyperparameter of a transformer. Give an example of a typical value of this hyperparameter from a recent open-weights model.

Solution.

The *model dimension* d_{model} is the length of vectors that represent input and output in the transformer. Intermediate computations or “states of mind” are also represented by vectors of this length. Example: The 2025 model Qwen3-32B has $d_{\text{model}} = 5120$. (An approximate number is accepted as correct.)

Question 3. Attention layer (5 points)

List the major quantities employed by an attention layer, in the order of computation.

Solution.

Masked self-attention, layer normalization, feed-forward network, layer normalization.

Question 4. Attention layer hyper-parameters (10 points)

List the two most important hyperparameters of an attention layer. Give a brief explanation of each, and a typical example from a recent open-weights model.

Solution.

- In the masked self-attention component, the *number of attention heads* H determines the *head dimension* $d_H = d_{\text{model}}/H$ (so H must be a divisor of d_{model}). We think of H and d_H as a single hyperparameter because the choice of one determines the other. Example: The 2025 model Qwen3-32B has $H = 64$ and thus $d_H = 5120/64 = 80$. (Approximate numbers are accepted as correct.)
- In the feed-forward network, the *FFN hidden dimension* d_{ff} is the number of units in the single hidden layer of the FFN. Example: The 2025 model Qwen3-32B has $d_{\text{ff}} = 5d_{\text{model}} = 25,600$. (An approximate number is accepted as correct.)

Question 5. Masked self-attention (5 points)

In the phrase *masked self-attention*, briefly describe the meaning of “masked” and “self”.

Solution.

“Masked” means that tokens cannot attend to future tokens—the future is *masked*. “Self” means that a token may attend to itself.

Question 6. Multi-head attention operator (5 points)

Describe the difference between single-head and a multi-head attention in one sentence.

Solution.

Multi-head attention splits the input representation by projecting it into H smaller representations, computing standard attention on each, then concatenating the results to produce the projected attention output.

Question 7. One-hot encoding (5 points)

Suppose we are using a vocabulary of 5 tokens, and we wish to encode the following sequence of three token IDs: $\text{seq} = (4, 2, 4)^\top$. What is the resulting one-hot token matrix?

Solution.

Token ID 4 encodes as $(0, 0, 0, 0, 1)$. Token ID 2 encodes as $(0, 0, 1, 0, 0)$. So the sequence of IDs $(4, 2, 4)^\top$ encodes as

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Question 8. Embedding and unembedding (5 points)

Describe the role of embedding and unembedding in a transformer model. Make sure to connect this to the values of the model dimension d_{model} , vocabulary size $|\mathcal{V}|$, and sequence length L . State the sizes of relevant matrices and vectors.

Solution.

The embedding transforms a sequence of one-hot token vectors into a representation in which each token is a vector of length d_{model} :

$$E_{L \times d_{\text{model}}} = T_{L \times |\mathcal{V}|} W_E_{|\mathcal{V}| \times d_{\text{model}}},$$

The unembedding transforms the output representation x_{last} of the final sequence position into logits:

$$\text{logits}_{\text{last}} = x_{\text{last}}_{1 \times d_{\text{model}}} W_U_{d_{\text{model}} \times |\mathcal{V}|}.$$

Question 9. Training targets (5 points)

A certain transformer uses lower case letters from the English alphabet as its only tokens. The following sequence is its only training data:

yymfndifhdjsjsjshfgkgjdjdvbkfrbsbivb

The transformer employs a block size of 5, and it is being trained using batches of size 2. One particular batch of training data employs the inputs

$$x = \begin{pmatrix} \text{hdjsj} \\ \text{jdvbk} \end{pmatrix}.$$

For this batch, what value of the targets y should be employed?

Solution.

We need sequences of the training data that match the inputs except shifted right by one position. This gives

$$y = \begin{pmatrix} \text{djsjs} \\ \text{dvbkf} \end{pmatrix}.$$

Question 10. Predictions and loss calculation

A transformer uses the vocabulary $\{\mathbf{a}, \mathbf{b}\}$ with IDs $\{\mathbf{a} : 0, \mathbf{b} : 1\}$. It processes a sequence of length 3, producing the final logits

$$\begin{pmatrix} 3.1 & -0.5 \\ 0.2 & 0.7 \\ -1.1 & -0.1 \end{pmatrix}.$$

(a) **(5 points)** What are the predicted probabilities for the next token, and which token will be output?

Solution.

Prediction uses logits for the final sequence position only, so we compute a softmax of $(-1.1, -0.1)$, yielding

$$\left(\frac{e^{-1.1}}{e^{-1.1} + e^{-0.1}}, \frac{e^{-0.1}}{e^{-1.1} + e^{-0.1}} \right) = (0.27, 0.73).$$

The highest probability corresponds to token ID 1, so the output is **b**.

(b) **(5 points)** Suppose instead that the same logits above were produced by training on a single input sequence $x = \mathbf{aba}$ and target $y = \mathbf{abb}$. What is the contribution of the final token to the cross-entropy loss?

Solution.

The total cross-entropy loss is $-\sum_i \log p_{y_i}$, where y_i is the token ID of the i th target token. The question asks for the contribution from the final token only. The final target is **b** (token ID 1). From the previous question, we know the computed probability of **b** was 0.73. So the contribution of this token to the loss is

$$-\log 0.73 = 0.14.$$

Question 11. Optional. Will not be on the quiz or exam. Single-head attention operator. (0 points)

List the major quantities employed by a single-head attention operator, in the approximate order of computation.

Solution.

token embedding, position embedding, input representation, query vectors, key vectors, value vectors, attention weights, attention-weighted value vectors, projected attention output, post-residual output representation.

See ?? for some optional additional explanation of this computation.

Appendix A Optional extra information about transformer computation

Information that appears only in this appendix is not required for quiz or exam questions.

- token embedding = one-hot token matrix $\cdot$ embedding matrix:

$$\underset{L \times d_{\text{model}}}{E_T} = \underset{L \times |\mathcal{V}|}{T} \underset{|\mathcal{V}| \times d_{\text{model}}}{W_E},$$

- position embedding = $\underset{L \times d_{\text{model}}}{E_P}$.
- input representation = token embedding + position embedding:

$$\underset{L \times d_{\text{model}}}{X} = \underset{L \times d_{\text{model}}}{E_T} + \underset{L \times d_{\text{model}}}{E_P}$$

- attention layers iteratively produce output representation X from input representation X :

$$\begin{aligned} X &\rightarrow \text{AttentionLayer}(X) \\ X &\rightarrow \text{AttentionLayer}(X) \\ &\vdots \\ X &\rightarrow \text{AttentionLayer}(X) \end{aligned}$$

- logits = output representation $\cdot$ unembedding matrix:

$$\underset{L \times |\mathcal{V}|}{\text{logits}} = \underset{L \times d_{\text{model}}}{X} \underset{d_{\text{model}} \times |\mathcal{V}|}{W_U}$$

- logits for last token = last row of logits:

$$\underset{1 \times |\mathcal{V}|}{\text{logits}_{\text{last}}} = \underset{L \times |\mathcal{V}|}{\text{logits}}(L - 1, :)$$

- probabilities for last token = softmax of logits for last token:

$$\underset{1 \times |\mathcal{V}|}{p} = \text{softmax} \left(\underset{1 \times |\mathcal{V}|}{\text{logits}_{\text{last}}} \right)$$

- predicted token ID i^* =
 - *either* the token ID with highest probability
 - *or* a token ID chosen randomly according to the calculated probabilities p ,

- *or* a token ID chosen randomly according to probabilities that have been adjusted via a temperature.

That is,

$$\begin{aligned}i^* &= \arg \max_i p_i, \text{ or} \\i^* &= \text{choice}(p), \text{ or} \\i^* &= \text{choice}(p').\end{aligned}$$

The adjusted probabilities p' , for a temperature $\tau > 0$, are given by

$$p'_i = \frac{p_i^{1/\tau}}{\sum_{j=0}^{L-1} p_j^{1/\tau}}.$$

Appendix B Optional extra information about single-head attention operator

Information that appears only in this appendix is not required for quiz or exam questions.

under construction